



# Predicting The Probability of Default for Banca Massiccia

LAVYA MIDHA & NIDHI DONURU

# Business Understanding

**The business problem:** Banca Massiccia wants to predict the probability of default of a potential borrower on a principal or interest payment for a prospective loan over the next 12 month.

Poor PD estimates → bad underwriting → higher expected losses, inefficient capital allocation, and regulatory non-compliance.

**Goal:** Build a data-driven, financially interpretable, and calibrated PD model to improve credit decisioning.

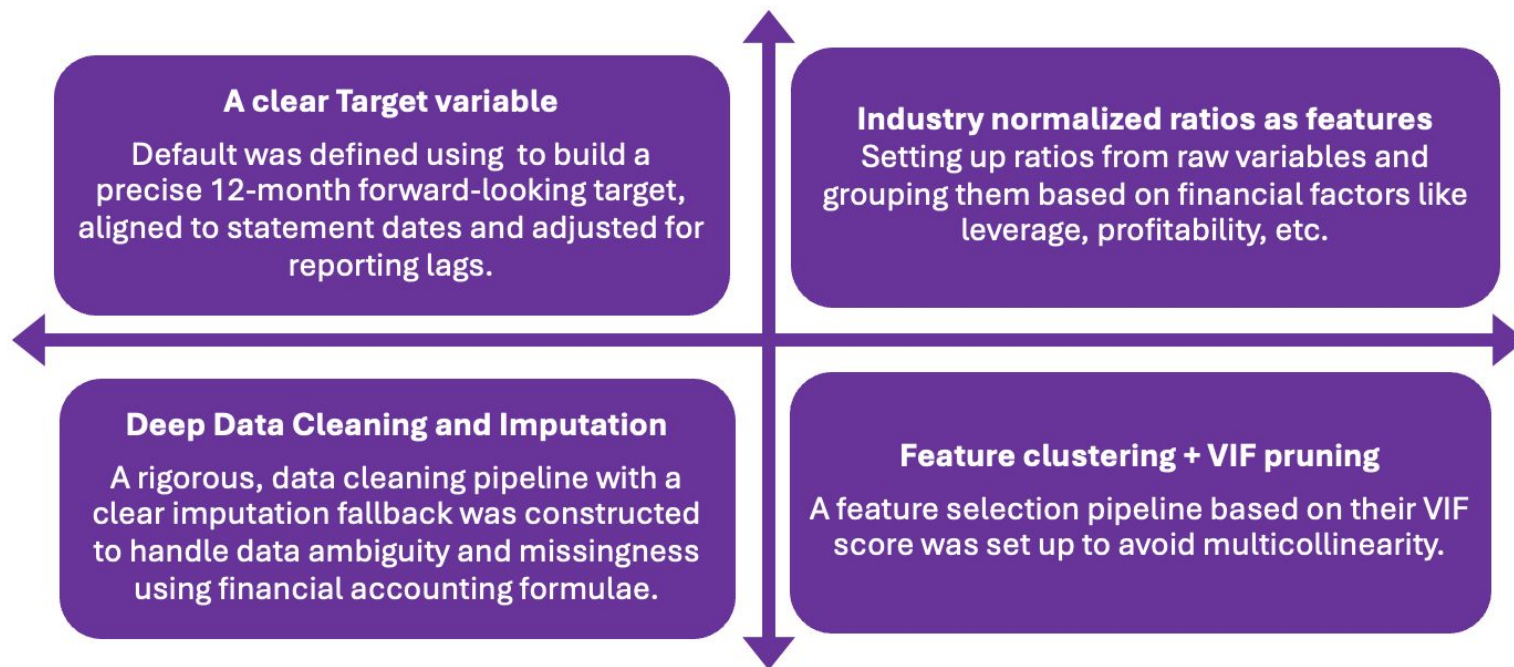
## How data mining addresses the business problem?

- Learns patterns from historical financial data and defaults to identify early signs of distress
- Captures non-linear financial distress behaviour (U-shaped ROE, cashflow, growth, ) that traditional linear-models cannot detect
- Models interactions between profitability, leverage, liquidity, size, cash-flow and growth ratios uncovering complex relationships
- Produces calibrated PDs aligned with observed default rates
- firm-level OOS and OOT splits to avoid leakage and replicate real-world conditions

# What has been done for the problem in the past ?

- **Subjective Assessment:** Initially, default risk assessments were based on subjective judgement, personal relationships and borrower's history
- **Quantitative Models (Altman z score):** A model that uses five financial ratios to predict a company's likelihood of default.
- **Logit/ Probit Models:** GLM based models that used a set of variables to predict the probability of default in a probabilistic framework
- **Finance Structural Models like Merton:** Merton's model provided a theoretical framework that linked a firm's default risk to its capital structure, treating the firm's equity as a call option on its assets
- **Modern Day ML Models:** Advanced AI and statistical techniques including neural networks to predict the probability of default

# Setting up the Problem with Financial Intuition



# Data Understanding

We used Banca Massiccia's dataset on their previous borrowers including information about the company type, industry, etc. as well as financial statement data on an annual basis for each borrower.

## Data Convention

Each row is one firm-year, with training sample including 1023552 rows.

Annual observations

44 variables in total

All quantities (except ratios) are reported in €

Only firms with > €1.5MM in assets are included in data

Only non-finance/insurance firms are included

There is no information in the training data regarding defaults that occurred after 12/31/2012

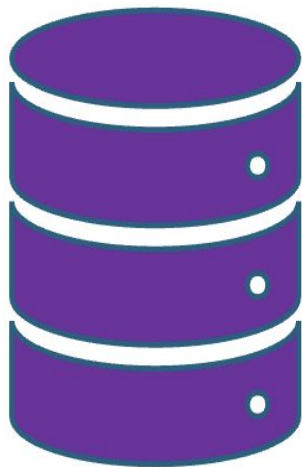
## Data Biases

Missing Data: Each row had at least one missing value with some columns like days rec being null in around 72.3% of the training data rows

Accounting Inconsistencies: A lot of the rows had violated general accounting inconsistencies like total short term bank debt > total short term bank debt

Missing or Irregular Years: Some firms had data missing between financial years, causing potential data imbalances

# Data Preparation



The following steps were followed for preparing data to be used for training our algorithm

- Data Cleaning
  - Filling in with accounting formulae
- Fallback mechanism via Cross-Sectional Approximate Factor Models
- Target Variable Engineering
- Feature Engineering
- Feature Selection

# Data Cleaning (A)

The following identities were used to fill in missing values and correct any accounting errors.

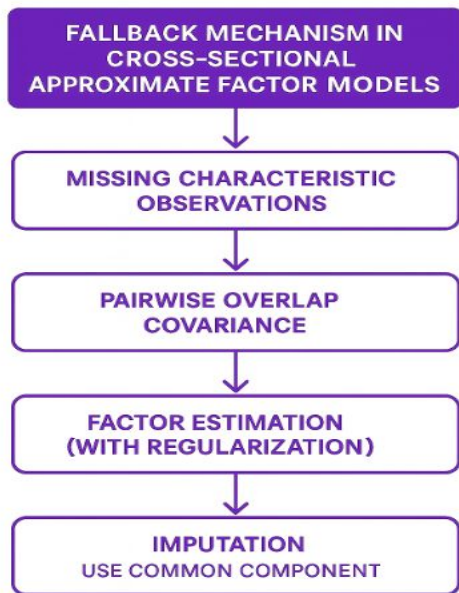
- $\text{debt\_st} = \text{debt\_bank\_st} + \text{debt\_fin\_st} + \text{AP\_st}$
- $\text{debt\_lt} = \text{debt\_bank\_lt} + \text{debt\_fin\_lt} + \text{AP\_lt}$
- $\text{asst\_tot} = \text{asst\_current} + \text{asst\_tang\_fixed} + \text{asst\_intang\_fixed} + \text{asst\_fixed\_fin}$
- $\text{Roa} = \text{profit} / \text{asst\_tot}$
- $\text{roe} = \text{profit} / \text{eqty\_tot}$
- $\text{rev\_operating} = \text{COGS} + \text{prof\_operations}$
- $\text{Margin\_fin} = \text{eqty\_tot} - \text{fixed\_assets}$
- $\text{Fixed\_assets} = \text{asst\_tang\_fixed} + \text{asst\_intang\_fixed} + \text{asst\_fixed\_fin}$
- $\text{Days\_rec} = (\text{AR} / \text{rev\_operating}) * 365$

## Quality Check Example

- A few identities were also used based on how much the overall data matches the formulae
- Example:  $\text{rev\_operating} = \text{COGS} + \text{prof\_operations}$  was used because it matched 992,057 out of the 1,023,270 rows.



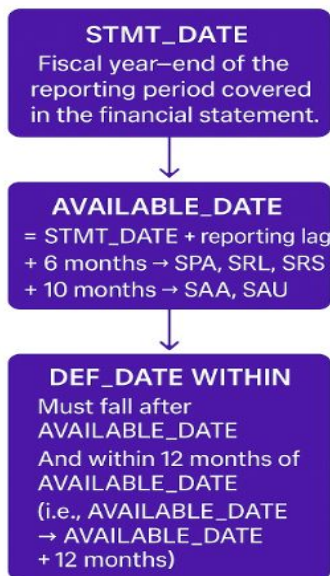
# Data Cleaning (B): Fallback



- The fallback mechanism is based on the paper, “Missing Financial Data” by Svetlana Bryzgalova, Sven Lerner, Martin Lettau and Markus Pelger. The authors guarantee state the technique to produce cleaner, realistic financial variables and reduce noise.
- After filling in using financial entities, the train data still had missing values. Dropping or median imputation seemed incorrect, instead the model falls back to **pairwise-available data** and **regularized factor estimation** to guarantee valid imputations under arbitrary missing patterns.
- After applying the fallback imputation, we run a final integrity check to ensure no accounting identities are violated. This step revalidates core relationships (e.g., assets, liabilities, equity) and corrects any inconsistencies, ensuring the dataset remains fully coherent.

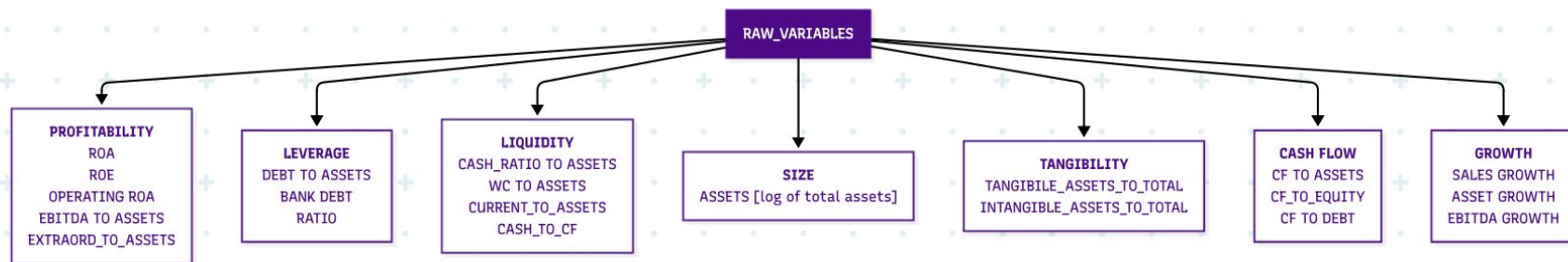


# Target Variable Engineering



- To train a supervised ML model, we engineered the target variable using each firm's `def_date`.
- Lag periods were determined by legal reporting requirements:
  - SPA, SRL, SRU, SRS: Italian corporations must publish financial statements within 180 days after fiscal year-end.
  - SAA, SAU: Firms without statutory reporting requirements were assigned a lag based on the tax-filing deadline (October 31), which requires all financial statements to be finalized.
- Using these lags, we computed an `available_date` and defined **default\_next\_year** = 1 if the firm's `def_date` falls within 12 months after the adjusted availability date, and 0 otherwise.

# Feature Engineering & Selection



- We used our raw variables, after being imputed for null values, to create financial ratios under 7 broad categories as the image shows
- We applied a group-wise feature selection procedure, choosing one representative variable from each financial category. For the chosen combination we calculated its VIF score to make sure a combination with low multicollinearity is selected.
- Our final feature set included 'roa', 'debt\_to\_equity', 'wc\_to\_assets', 'assets', 'tangibility',

# Modeling

## Approaches explored

### Altman Z-score

#### Pros:

- Simple, fast, interpretable
- Works with limited data
- No training required

#### Cons:

- Fails on U-shaped relationships (ROE, growth, cashflow)
- Cannot adapt to new data and patterns
- Profitability x leverage x liquidity effects are ignored

### Logistic regression

#### Pros:

- Highly interpretable
- Supports WOE binning
- Works well with monotonic ratios

#### Cons:

- Assumes monotonic relations
- Cannot learn non-linear patterns or interactions
- Requires heavy preprocessing (binning, WoE, transformations)

### XGBoost

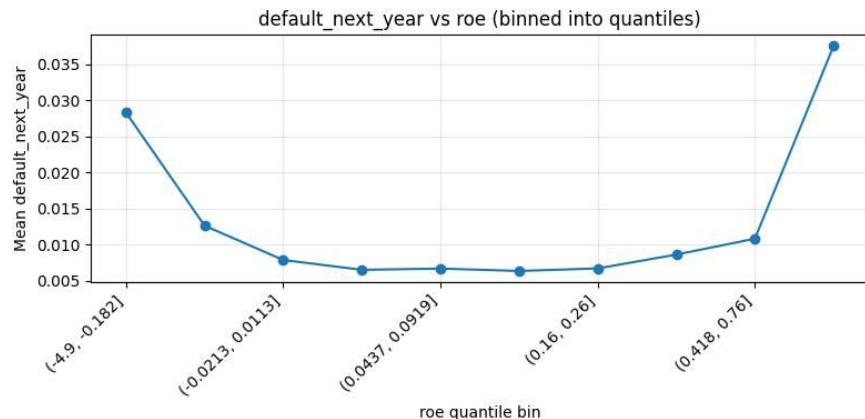
#### Pros:

- Interprets non-linear patterns
- Models interactions automatically (size x growth)
- Learns around noise (accounts for 0s and outliers)
- Strongest predictive accuracy in literature

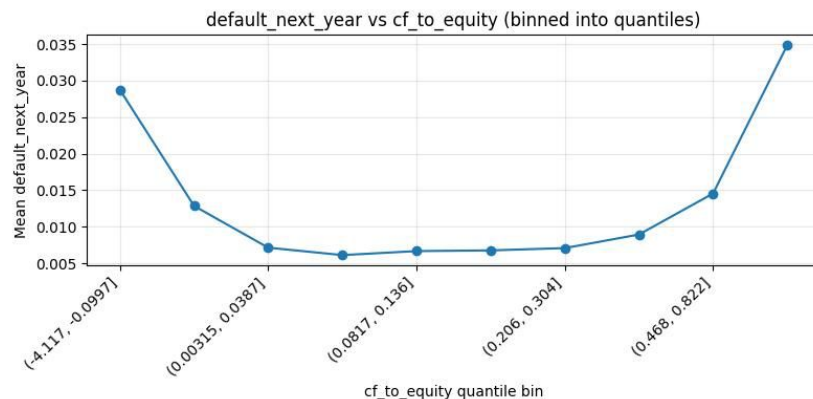
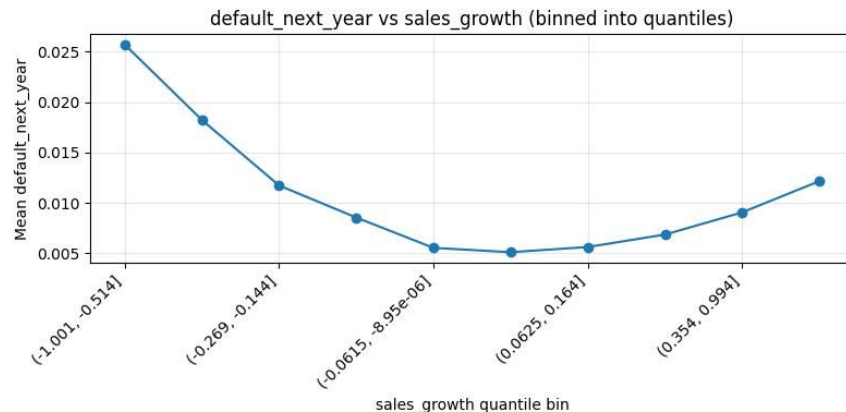
#### Cons:

- Less interpretable than logistic and altman
- Needs hyperparameter tuning
- Not naturally calibrated PDs
- Computationally heavier

# A few mined patterns w.r.t default



- Double-edged swords - extreme low and high values increase default probability
- Manual binning is possible but introduces arbitrary cutoffs and biases
- Tree-based models capture these patterns naturally without imposing monotonic assumptions



# Final Model

We went with XGB with the for our final model. This is how our model looks like:

```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
               colsample_bylevel=None, colsample_bynode=None,
               colsample_bytree=0.8, device=None, early_stopping_rounds=None,
               enable_categorical=False, eval_metric='logloss',
               feature_types=None, feature_weights=None, gamma=None,
               grow_policy=None, importance_type=None,
               interaction_constraints=None, learning_rate=0.05, max_bin=None,
               max_cat_threshold=None, max_cat_to_onehot=None,
               max_delta_step=None, max_depth=4, max_leaves=None,
               min_child_weight=None, missing=nan, monotone_constraints=None,
               multi_strategy=None, n_estimators=600, n_jobs=-1,
```

# Evaluation: Train-test Split

## Training set:

All firms with financial statements  $\leq 2010$ , excluding the firms held out for OOS.

Out-of-sample (OOS) validation: A random 20% of firms from  $\leq 2010$  held out for calibration, ensuring the calibration is performed on unseen borrowers, not just unseen rows.

Out-of-time (OOT) test: All observations with statement year  $\geq 2011$ , ensuring performance is measured on future years to mimic real underwriting.

# Calibration

**Objective: Align raw model-predicted PDs with empirical default probabilities, ensuring interpretability, stability.**

- Get raw PDs on Out of Sample data set using XGBoost.
- Sort loans by raw PDs and split it into 20 buckets (with at least 300 observations in each bucket)
- For each bucket compute:
  - Mean raw PD
  - Empirical default rate using Laplace smoothing
- Map Raw  $\rightarrow$  Calibrated PD
- Test Calibration on OOT Data
  - Locate its raw PD bucket
  - Assign the bucket's calibrated PD



# Metrics and Stats

## Kolmogorov-Smirnov (KS) statistic

- Indicates how well the model separates good vs. bad borrowers

Range:

- <20, it is considered as a weak model.
- 20-40, it is considered acceptable.
- 40-70, it is considered as a good model.

## Brier Score

- Measures the mean squared error between predicted probabilities and actual outcomes.

Range:

- 1, perfect inaccuracy
- 0, perfect accuracy

Lower Brier Score: better calibrated and more accurate PDs.

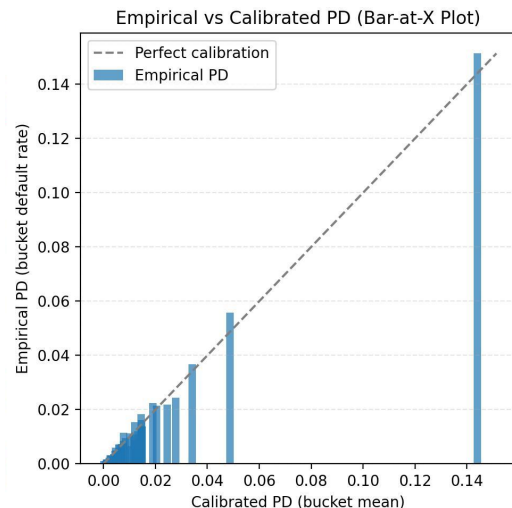
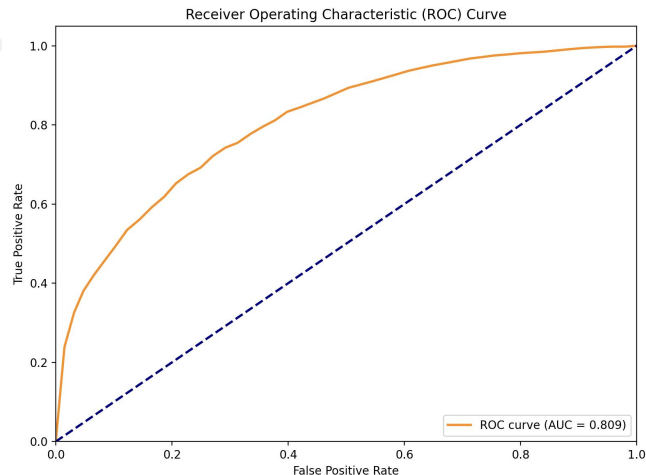
**NOTE:** Making a model better calibrated doesn't necessarily make it more powerful, therefore AUC remains similar after calibration.

## Our Metrics

AUC: 0.8087

BRIER: 0.0105

KS: 0.4511



# How a business case should be developed to project expected improvement? ROI?

It's not possible to compute ROI or financial gain because we don't have

- Loan information
- Interest rates
- Loss given default
- Exposure at default

$\text{ROI} = \text{Revenue from loan} - \text{Expected Loss} - \text{Capital Cost}$

The model is designed to estimate a borrower's probability of default; it is not intended to project the return on investment of making the loan.

# Deployment

The model is deployed into the overall workflow as follows



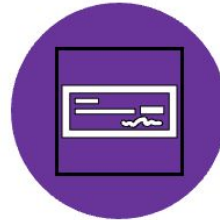
A potential buyer applies for a loan by providing information about the finances of the firm in the form of financial statement data



The data is fed into the model to predict how likely it is that the borrower will default.



The bank officer then determines what the appropriate interest rate and underwriting fees would be for the loan to be profitable, in consideration of the default probability.



If the bank officer concludes that the loan may be made at a rate that is within the bank's guidelines, the loan is made.

# Risks Associated with Deployment

- **Scope Limitation:** The model must not be applied to firms with < €1.5MM in assets and to finance/insurance firms, because the training data only included firms above this threshold and non - finance/insurance firms. Predictions outside this domain are unreliable.
- **Imputation & Sampling Bias Risk:** The fallback imputation strategy uses latent-factor–based similarity across firms to fill missing values, which may introduce sampling bias in segments where comparable firms are sparse or unrepresentative.
- **Human-in-the-Loop Requirement:** Credit officers must remain the ultimate decision-makers. PD is a decision-support tool, not a standalone approval/rejection system. Overrides should be allowed with documented justification.
- **Ongoing Monitoring & Backtesting:** Regular back-testing, drift monitoring, and annual recalibration are required to ensure the model remains stable, compliant, and aligned with portfolio-level PDs.

# Resources

**Agenzia delle Entrate.** (2025, May 16). *How and when to file a tax return.* <https://www.agenziaentrate.gov.it/portale/web/english/how-and-when-to-file-a-tax-return1>

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# Resources

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**Nikhil, D.** (2025). *Kolmogorov–Smirnov statistic explained: Measuring model power in credit risk modeling*. Towards Data Science.

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<https://towardsdatascience.com/laplace-smoothing-in-naive-bayes-algorithm-9c237a8bdece>

**Square Inc.** (2019). *Brier Score – PySurvival*. Retrieved from

[https://square.github.io/pysurvival/metrics/brier\\_score.html](https://square.github.io/pysurvival/metrics/brier_score.html)

# Appendix : Contributions

## Nidhi Donuru

1. Data cleaning - if data is behaving acc to financial ratios and formulae. What should be an assumption, what should be a true value
2. Finding references for legal structs SAA, SAU (stato patrimoniale)
3. Feature engineering for growth ratios, size
4. How features move w.r.t default rate
5. Model selection, model training

## Lavya Midha

1. Data Cleaning (Filling with financial identities and fallback)
2. Target Variable Engineering (including finding references for legal structures and their lag periods for all firms)
3. Feature Engineering for leverage, liquidity, profitability, cashflow
4. Feature Selection via VIF
5. Calibration



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# THANKS!

## Q&A

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